Propositional Logic

A **proposition** is a declarative statement that can either be *true* or *false*.

Example 1.1 :

- *Earth is a planet.*
- *Sun is a star.*

Both of the above sentences are propositions, and their truth value is **True**. The propositions in Example 1.1 cannot be broken down into simpler propositions. We will say they are ***elementary*** or ***atomic*** propositions.

A proposition can take an *affirmative* or a *negative* form.

Example 1.2:

- *Earth is round. (affirmative form)*
- *Earth is not round. (negative form)*

A proposition can also be the composition of several elementary propositions. The truth value of the entire proposition is determined by the truth value of each of the elementary propositions.

Example 1.3:

- *Sun rises in the east and sets in the west.*

This proposition is true because it is the conjunction of two propositions that are both true: "*Sun rises in the east*" and "*Sun sets in the west*." The proposition "*Earth is not round*" is not atomic. It is the negation of the atomic proposition "*Earth is round*."

Paradoxes. Declarative statements that we cannot decide their truth or falsity.

Example 1.4:

- *This sentence is false.*

If the statement is true (the sentence is false), its negation (this sentence is not false) is true. Therefore, the sentence is not false. But if it is true, it must be false, and thus true. We cannot say whether the sentence is true or false.

1.1. Propositional Language

A language, whether formal or natural, is characterized by its *syntax* and *semantics*.

1.2.1. The Syntax of Propositional Language

The formulas are written with the symbols of an alphabet according to well-defined writing rules.

1.1.1.1 The alphabet

The alphabet of propositional language includes :

- A set of propositional variables, which we will designate by uppercase letters, possibly

indexed, from the Latin alphabet (P, Q, P_1, P_2, ...).
- Logical connectives: ⌉, ∧,→,↔,→.
- Parentheses: "(", ")".

1.1.1.2 Writing Rules

In the following rules and throughout the rest of the course, we will agree to denote formulas using lowercase letters from the Greek alphabet.

r1. Every propositional variable is a (atomic) formula.
r2. If α is a formula, then (α) is a formula.
r3. If α is a formula, then⌉α is a formula.
r4. If α and βare two formulas, then the expressions: α∧β, α∨β, α→ β and α↔ β are formulas.
r5. No other expression is a formula of the language..

Definitions.
- An *atomic formula* refers to a propositional variable.
- A *literal* refers to an atomic formula or the negation of an atomic formula (P and ⌉P, for example).
- Two literals are complementary if one is the negation of the other.

1.1.2 Precedence of Connectives

Rules of precedence between connectives enables a unique reading and, thus, prevents ambiguities. In a given formula, we will apply the connectives in the following order: ⌉,∧, ∨, →, ↔. We will apply them from left to right when they appear consecutively in a formula.

Examples :

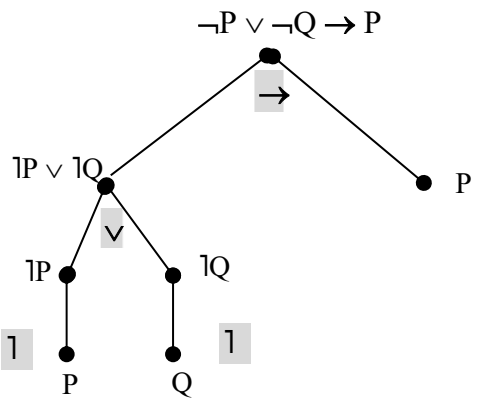
P → Q → R	is read as	(P → Q) →R
P → Q →R→ S	is read as	((P →Q)→ R)→ S
P ∨ Q ∨ R	is read as	(P∨Q) ∨ R
P ∧ Q ∨ R	is read as	(P∧ Q) ∨ R

Sub-formulas

The notion of sub-formula can be deduced from the writing rules described in section 1.2.1.2. The set S(α) of sub-formulas of a formula α can be defined as follows:

r1. If α is a formula then α∈ S(α)
r2. If α=⌉β is a formula then β∈ S(α)
r3. If α=β_1o β_2 is a formula then β_1, β_2∈ S(α) / o ∈ {∧, ∨, →, ↔}.
r4. If β∈ S(α) the S(β)⊂ S(α)

Example 1.5:
α **:** ⌉P ∨⌉Q→ P
S(α) ={ ⌉P ∨⌉Q→ P, ⌉P ∨⌉Q , P,⌉P, ⌉Q, Q }
The tree in Figure 1.3.1 shows the decomposition into sub-formulas of a given formula.



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1.2. Semantics of Propositional Logic

For a propositional language L, the purpose of semantics is to assign a truth value to the formulas of L. To do this, it associates a truth function or valuation function v: VP →{T,F}: to every formula α in L, where n denotes the number of propositional variables appearing in the formula. The function v can be defined as follows:

$v(P)$ = V or F for any propositional variable P in PV

$$v(\neg\beta) = \begin{cases} V \text{ if } v(\beta) = F \\ F \text{ if } v(\beta) = T \end{cases}$$

$$v(\alpha\wedge\beta) = \begin{cases} V \text{ if } v(\alpha) = T \text{ and } v(\beta) = T \\ F \text{ if } v(\alpha) = F \text{ or } v(\beta) = F \end{cases}$$

$$v(\alpha\vee\beta) = \begin{cases} V \text{ if } v(\alpha) = T \text{ or } v(\beta) = T \\ F \text{ if } v(\alpha) = F \text{ and } v(\beta) = F \end{cases}$$

$$v(\alpha\rightarrow\beta) = \begin{cases} V \text{ if } v(\alpha) = F \text{ or } v(\beta) = T \\ F \text{ if } v(\alpha) = T \text{ and } v(\beta) = F \end{cases}$$

$$v(\alpha\leftrightarrow\beta) = \begin{cases} V \text{ if } v(\alpha) = v(\beta) \\ F \text{ else} \end{cases}$$

To a formula α in which appear n distinct propositional variables, there correspond2^n valuation functions, which we can represent in a truth table with 2^n rows, where row i indicates the truth value of α for the valuation v_i. Tables 1.1 and 1.2 show the truth function of the connectives ¬, ∧, ∨, →, ↔.

V(P)	V(¬P)
T	F
F	T

Table 1.1

v(P)	v(Q)	v(P∧Q)	v(P∨Q)	v(P→Q)	v(P ↔Q)
T	T	T	T	T	T
T	F	F	V	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Table 1.2

As an example, Table 1.3 describes the truth functions of the formula P∨Q→R. We will omit writing v to simplify the notation.

P	Q	R	P ∨Q → R
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	T

Table 1.3.

In some situations, we do not need the truth values of all the propositional variables that appear in a formula α to determine whether its truth value is T or F. As an example, the truth value of P∨Q→R is T whenever v(R)=T. *v* is then called a *partial valuation function.*

1.2.1. Satisfiability of a Formula, a Set of Formulas

A valuation *v* satisfies a formula α if, after assigning the truth values associated with *v* to the propositional variables of α, the truth value of α is T ($v(\alpha)$ = T). We will note this as $v \models \alpha$. We will also write $v \not\models \alpha$ to mean « *v* does not satisfy *α* » or « *v* falsifies *α* ». The valuations that satisfy a formula α correspond to the rows of the truth table where the truth value of α is T. As examples:

- The valuation v_1(P) = T and v_1(Q) = T which corresponds to the first row of Table 1.2 , satisfies the formulas *P∧Q*, P∨Q, P→Q and P↔Q ;
- The valuation v_2(P) = T, v_2(Q) = F satisfy the formula P∨Q but does not satisfy the formulas P∧Q, P→Q and P↔Q.

A formulaαis satisfiable if there exists at least one valuation *v* such that $v \models \alpha$. α is said to be unsatisfiable otherwise.

Proposition 1

If $\beta_1$and β_2 are two formulas and *v* is a valuation then :

- For every propositional variable P, if v(P) = T then $v \models P$
- $v \models \neg\beta$ *if and only if* $v \not\models \beta$
- $v \models \beta_1 \wedge \beta_2$ *if and only if* ($v \models \beta_1$ and $v \models \beta_2$)
- $v \models \beta_1 \vee \beta_2$ *if and only if* ($v \models \beta_1$or $v \models \beta_2$)
- $v \models \beta_1 \rightarrow \beta_2$ *if and only if* ($v \not\models \beta_1$ or $v \models \beta_2$)
- $v \models \beta_1 \leftrightarrow \beta_2$ *if and only if* ($v \models \beta_1 \rightarrow \beta_2$ and $v \models \beta_2 \rightarrow \beta_1$)

We generalize the notion of satisfiability to a set of formulas: A set of formulas[1] de formulas Γ = { α_1, ..., α_n} is satisfiable, if and only if, there exists at least one valuation *v* that satisfies all the formulas inΓ ($v \models \alpha_1$, $v \models \alpha_2$, ... and $v \models \alpha_n$**).** In this case**,** *v* corresponds to a row in the truth table where the truth values of α_1, ..., α_n are all T.

Example 1.8 :

The set { P ∧ Q, P ∨ Q, P → Q, P ↔ Q } is satisfiable. To verify this, consider the valuation v_1

[1]We will denote sets of formulas by uppercase Greek alphabet letter.

corresponding to the first row of Table 1.4. In contrast, the set { $P \wedge Q$, $P \vee Q$, $P \rightarrow Q$, $P \leftrightarrow Q$, $\neg P$} is unsatisfiable (tableau 1.4).

P	Q	P∧Q	P ∨Q	P →Q	P↔Q	¬P
T	T	T	T	T	T	F
T	F	F	T	F	F	F
F	T	F	T	T	F	T
F	F	F	F	T	T	T

Table 1.4

1.1.3 Tautology

A formula α is a *tautology* (denoted|= α) if and only if *v*|= αfor every valuation *v*.

Example 1.10 :

|= P∧Q →Q (table 1.5).

P	Q	P∧Q	P∧Q→Q
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Table 1.5

1.1.4 Logically Equivalent Formulas

Two formulas α and β are logically equivalents *if and only if* $v(\alpha)= v(\beta)$ for every valuation *v*, in which case |=α↔ β.

Properties of Logical Connectives

Commutativity of∨ and ∧

$$\alpha\vee\beta \equiv\beta\vee\alpha \qquad \alpha\wedge\beta\equiv\beta\wedge\alpha$$

Associativity of ∨and ∧

$$(\alpha\vee\beta)\vee\gamma \equiv\alpha\vee(\beta \vee\gamma) \qquad (\alpha\wedge\beta)\wedge \gamma\equiv\alpha\wedge(\beta \wedge\gamma)$$

Distributivity of∨ and ∧

$$(\alpha\vee\beta)\wedge \gamma\equiv(\alpha\vee \gamma)\wedge(\beta\vee \gamma) \qquad (\alpha\wedge\beta) \vee \gamma\equiv(\alpha\vee \gamma)\wedge (\beta \vee \gamma)$$

Idempotence

$$\alpha\vee\alpha\equiv\alpha \qquad \alpha\wedge\alpha\equiv\alpha$$

Loi de De Morgan

$$\neg(\alpha\vee\beta)\equiv\neg\alpha\wedge \neg\beta \qquad \neg(\alpha\wedge\beta)\equiv\neg\alpha\vee \neg\beta$$

$$\neg\neg\alpha\equiv\alpha$$

1.2.2. Logical Consequence

A formula β is a logical consequence of a formula α (denoted α|= β), if and only if, for every valuation *v*; *if v*|=α *then v*|= β.

On the truth table of α and β, the truth value of β is Ton all the rows where the truth value of α is T.

In general, a formula β is a logical consequence of a set of formulas Γ : $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ (denoted $\alpha_1, \ldots, \alpha_n \models \beta$ more simply $\Gamma \models \beta$) if and only if:

For all valuations *v*, if $v \models \alpha_1$ and $v \models \alpha_2$ and ….. and $v \models \alpha_n$ then $v \models \beta$

Example 1.6:
$P \rightarrow Q, \neg Q \models \neg P$ (Table 1.6).
In contrast, ⌉Q is not a logical consequence of {P → Q, ⌉P} ($P \rightarrow Q, \neg Q \not\models \neg Q$). The valuation corresponding to row 3 of Table 1.6 satisfies the formulas P → Q, ⌉P but does not satisfy ⌉Q.

P	**Q**	**⌉Q**	**P →Q**	**⌉P**
T	T	F	T	F
T	F	T	F	F
F	T	F	T	T
F	F	T	T	T

Table 1.6

Example 1.7:

$P \wedge Q \rightarrow R, P \wedge \neg R \models Q \rightarrow R$ (Table 1.7).

P	**Q**	**R**	**P ∧ Q → R**	**P∧⌉R**	**Q→ R**
T	T	T	T	F	T
T	T	F	F	T	F
T	F	T	T	F	T
T	F	F	**T**	**T**	**T**
F	T	T	T	F	T
F	T	F	T	F	F
F	F	T	T	F	T
F	F	F	T	F	T

Table 1.7

Theorem 1.1.

$\Gamma \models \beta$ *if and only if* $\Gamma \cup \{\neg\beta\}$ unsatisfiable

Demonstration

⇒*If* $\Gamma \models \beta$ *then* the set $\Gamma \cup \{\neg\beta\}$ is unsatisfiable
Let us suppose (1) that $\Gamma \models \beta$ and (2) that the set $\Gamma \cup \{\neg\beta\}$ is satisfiable. Then, there would exist at least one valuation (assumption 2) such that v satisfies Γ and ⌉β. (i.e $v \not\models \beta$). This contradicts assumption 1.
⇐*If* $\Gamma \cup \{\neg\beta\}$ is unsatisfiable *then* $\Gamma \models \beta$.

Let us suppose (1) that : $\Gamma \cup \{\neg\beta\}$ is unsatisfiable and (2)that: $\Gamma \not\models \beta$.There would exist then (by assumption 2) at least one valuation v that satisfies $\Gamma$and $v \models \neg\beta$. In this case, the set $\Gamma \cup \{\neg\beta\}$ would be satisfiable, which contradicts assumption 1.

Particular case.

$\models \beta$ *if and only if* $\neg\beta$ unsatisfiable.

Theorem 1.2.

$\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ *if and only if* $\alpha_1, \alpha_2, \ldots, \alpha_{n-1} \models \alpha_n \rightarrow \beta$

Particular case **(***n* = 1)

$\alpha \models \beta$ *if and only if* $\models \alpha \rightarrow \beta$

The left-hand side of the theorem indicates that β is a logical consequence of the set : $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$and the right-hand side indicates that $\alpha_n \rightarrow \beta$ is a logical consequence of the set : $\{\alpha_1, \alpha_2, \ldots, \alpha_{n-1}\}$

Proof (by contradiction):

If $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ *then* $\alpha_1, \alpha_2, \ldots, \alpha_{n-1} \models \alpha_n \rightarrow \beta$

Suppose that the right-hand side of the proposition is false (i-e $\alpha_1, \alpha_2, \ldots, \alpha_n \not\models \beta$ Then, there would exist at least one valuation v such that :

$v \models \alpha_1$ and $v \models \alpha_2$ and ….. and $v \models \alpha_{n-1}$ and such that $v \models \alpha_n \wedge \neg\beta$, thus $v \models \alpha_n$ and $v \models \neg\beta$ (proposition 1)

The valuation v would therefore satisfy all the formulas $\alpha_1, \alpha_2, \ldots, \alpha_n$ but not the formula β. This would contradict the assumption (left-hand side).

Corollary 1.1.

$\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ *if and only if* $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow \beta)))..)$

Demonstration (by induction on *n*)

i. *n = 1.* (Particular case of theorem 1.2).

Inductive Hypothesis (assuming the corollary for order n)

$\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ *if and only if* $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow \beta)))..)$

ii. *For order n+1.*

⇒) Replace β by $\alpha_{n+1} \rightarrow \beta'$ in the inductive hypothesis

If $\alpha_1, \alpha_2, \ldots, \alpha_n \models \alpha_{n+1} \rightarrow \beta'$ *then* $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow (\alpha_{n+1} \rightarrow \beta')))..)$ **(1)**

If $\alpha_1, \alpha_2, \ldots, \alpha_n, \alpha_{n+1} \models \beta'$ *then* $\alpha_1, \alpha_2, \ldots, \alpha_n \models \alpha_{n+1} \rightarrow \beta'$(theorem 1.1) **(2)**

If $\alpha_1, \alpha_2, \ldots, \alpha_n, \alpha_{n+1} \models \beta'$ *then* $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow (\alpha_{n+1} \rightarrow \beta')))..)$

(Transitivity between 2 et 1)

⇐) *If* $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow \beta)))..)$ *then* $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ (Inductive Hypothesis).

By replacing β by $\alpha_{n+1} \rightarrow \beta'$ we obtain:

If $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow (\alpha_{n+1} \rightarrow \beta')))..)$ *then* $\alpha_1, \alpha_2, \ldots, \alpha_n \models \alpha_{n+1} \rightarrow \beta'$**(1')**

If $\alpha_1, \alpha_2, \ldots, \alpha_n \models \alpha_{n+1} \rightarrow \beta'$ *then* $\alpha_1, \alpha_2, \ldots, \alpha_n, \alpha_{n+1} \models \beta'$ (theorem 1.1) **(2')**

if $\models \alpha_1 \rightarrow (\alpha_2 \rightarrow \ldots\ldots \rightarrow (\alpha_{n-1} \rightarrow (\alpha_n \rightarrow (\alpha_{n+1} \rightarrow \beta')))..)$ *then* $\alpha_1, \alpha_2, \ldots\ldots, \alpha_n, \alpha_{n+1} \models \beta'$
(Transitivity between 2' et 1')

Theorem 1.3.

If $\Gamma \models \alpha$ *and* $\Gamma \models \alpha \rightarrow \beta$ *then* $\Gamma \models \beta$

Particular Case($\Gamma = \varnothing$)

If $\models \alpha$ *and* $\models \alpha \rightarrow \beta$ *then* $\models \beta$

Proof in the exercise.

Theorem 1.4.

If $\Gamma \models \beta$ *and* $\Gamma \models \neg\beta$ *then* Γ is unsatisfiable.

Proof

Suppose that $\Gamma \models \beta$ (hyp. 1) and $\Gamma \models \neg\beta$ (hyp. 2) and that Γ is satisfiable. Then, there would exist a valuation v that satisfies all the formulas in Γ. The same valuation v would also satisfy β($v \models \beta$) according to Hypothesis 1 and satisfies $\neg\beta$ ($v \models \neg\beta$) according to Hypothesis 1. This is, of course, impossible.

Theorem 1.5. *(Substitution Theorem)*

Let β be a formula in which the propositional variable P appears in one or more *occurrences,* and let β' be the formula obtained from β by substituting the formula α <u>for each occurrence</u> of *P then*, *If* $\models \beta$ *then* $\models \beta'$.

Example 1.8 :

$\beta : P \rightarrow ((Q \rightarrow R) \rightarrow P)$. (It is easily verified that β is a tautology.)

β': $(A \wedge B) \rightarrow ((Q \rightarrow (B \vee C)) \rightarrow (A \wedge B))$ obtained by substituting $(A \wedge B)$ for P and $(B \vee C)$ for R in β. It is equally easy to verify that β' is a tautology.

Theorem 1.6.*(replacement theorem)*

Let α be a formula in which the formula β appears one or more times (i.e., α is of the form: ..... β ..... β..... β.....). If we denote by α' the formula obtained from α by replacing β <u>in one or more of these occurrences</u> with a formula β', then

If $\models \beta' \leftrightarrow \beta$ *then* $\models \alpha' \leftrightarrow \alpha$.

Example 1.9:

$\alpha : P_1 \vee (P_2 \rightarrow P_3) \leftrightarrow P_4 \vee (P_2 \rightarrow P_3)$

By replacing the first occurrence of $(P_2 \rightarrow P_3)$ with $(\neg P_2 \vee P_3)$ we obtain the formulaα': $P_1 \vee (\neg P_2 \vee P_3) \leftrightarrow P_4 \vee (P_2 \rightarrow P_3)$ which is logically equivalent toα.

1.2.3. Complete Set of Connectives

Definition 1.1. A set S of connectives is complete if for every formula α in the language L, there exists a formula α' in which only the connectives from S appear, and such that $\alpha' \equiv \alpha$.

Example 1.10:

the $\{\neg, \wedge\}$ forms a complete system.

Sheffer Connectives

Proposition. There are two complete sets of connectives, each formed by a single connective.

1.2.4. Disjunctive Normal Form and Conjunctive Normal Form:

Definition 1.2. A formula α is in Disjunctive Normal Form (DNF) if it is of the form $M_1 \vee M_2 \vee \ldots \vee M_n$ where each M_i is of the form $L_1 \wedge L_2 \wedge \ldots \wedge L_n$ and every L_i represents either the literal P_i or the literal $\neg P_i$.

Example 1.10 :

$P \vee (Q \wedge R) \vee (\neg Q \wedge \neg P)$

Definition 1.3. A formula α is in Conjunctive Normal Form (CNF) if it is of the form $C_1 \wedge C_2 \wedge \ldots \wedge C_n$ where each C_i is of the form $L_1 \vee L_2 \vee \ldots \vee L_n$.

Example 1.11 :

$(P \vee Q \vee R) \wedge (\neg Q \vee \neg P) \wedge (Q \vee \neg P)$

Theorem 1.7. For every formula α, there exists a formula α′ in DNF (or CNF) such that:

$$\models \alpha' \leftrightarrow \alpha.$$